VQA-DISAGREE: Multi-Model Disagreement as an Annotation-Free Difficulty Signal for VQA Benchmarks*

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Abstract

Visual question answering benchmarks treat all samples as equal-weight evaluation signal, yet in practice a large fraction are trivially easy: every current vision-language model (VLM) answers them correctly. Such saturated *samples contribute almost no information when comparing models. We propose* multi-model disagreement*—the fraction of distinct answer clusters produced by an ensemble of four architecturally diverse VLMs—as a cheap, annotation-free proxy for sample difficulty. Applying this metric to 1,500 questions drawn from four public VQA benchmarks spanning natural scenes, scene-text, charts, and real-world photographs (VQAv2, TextVQA, ChartQA, and RealWorldQA), we find that* ***5.4%*** *of samples are saturated (unanimous and correct), while* ***50.1%*** *are genuinely hard (high-disagreement). Qwen2-VL-72B used as a scale-based oracle confirms a* ***28.3*** *pp Easy–Hard accuracy gap and an even larger* ***36.6*** *pp Easy–Deceptive gap (38.3% vs. 1.7%), validating disagreement as a difficulty signal. The* deceptive *subcategory—unanimous but wrong—turns out to be the hardest stratum for the oracle, suggesting it captures shared architectural blind spots rather than mere model-family idiosyncrasies. We release* VQA-DISAGREE*, a 456-sample difficulty-stratified subset on HuggingFace Hub, intended as a more informative evaluation slice for the multimodal community.*

1. Introduction

The rapid capability growth of vision-language models has created a familiar problem in benchmark evaluation: saturation. On established datasets such as VQAv2 [5] and GQA [6], top-performing models already exceed 80% accuracy, compressing the signal available for distinguishing them. Comprehensive evaluations confirm this trend across dozens of multimodal tasks [8]. Yet benchmark curators rarely retire samples once they become easy, meaning that each new paper wastes compute evaluating questions that every VLM has already solved.

The standard remedy—collecting new, harder benchmarks—is expensive and slow. We argue for a complementary, automatic approach: rather than replacing benchmarks, *stratify* them by difficulty using a signal that is already available: the *disagreement* of a small ensemble of diverse VLMs [13]. When four models of different architectures all give the same answer, that question is very likely easy (or at worst, consistently misleading). When they give four different answers, the question is genuinely ambiguous or hard. Rather than discarding or replacing saturated benchmarks, this annotation-free stratification automatically identifies the subset that still provides discriminative signal.

Concretely, we define a disagreement score $D \in [0, 1]$ computed from semantic clustering of predictions (Section 3) and use it to assign each sample to one of four strata: *Easy* ($D = 0$, all correct), *Deceptive* ($D = 0$, all wrong), *Medium* ($D = 0.25$), and *Hard* ($D \geq 0.5$). We apply this pipeline to 1,500 samples spanning four publicly available benchmarks, validate the strata against a Qwen2-VL-72B oracle, and release VQA-DISAGREE—a 456-sample difficulty-stratified subset—on HuggingFace Hub.

Contributions.

1. A reproducible, annotation-free difficulty metric based on multi-model semantic disagreement (§3).
2. An empirical characterisation of how difficulty distributes across four public VQA benchmarks and seven question types (§4).
3. Validation that both high-disagreement and unanimous-but-wrong samples are genuinely harder, using Qwen2-VL-72B as a scale-based oracle on 455 items spanning all four strata (§4.3).
4. The VQA-DISAGREE dataset: 456 difficulty-stratified VQA samples with full metadata, released publicly for community use (§5).

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2. Related Work

Benchmark saturation and renewal. Benchmark saturation—where top models converge near ceiling accuracy—is now well-documented in vision-language evaluation [8]. Responses include constructing harder benchmarks and data cartography [13], which uses training dynamics to identify *easy*, *ambiguous*, and *hard-to-learn* samples. Our work adapts the cartography intuition to the multimodal, inference-only setting: we replace training dynamics with multi-model disagreement, requiring no fine-tuning.

Data quality and pruning. Prior work on dataset pruning [12] and quality-guided selection [4] shows that a well-chosen subset outperforms random sampling at equal size. We bring this lens to VQA *evaluation*: our disagreement score identifies the subset of existing benchmarks that still provides discriminative signal.

VLM evaluation benchmarks. Recent work has exposed systematic visual shortcomings of current VLMs [14], including failures on chart interpretation and fine-grained visual grounding. Our contribution is orthogonal: rather than curating new content, we characterise difficulty within *existing* benchmarks and release a stratified slice that concentrates signal in the Hard and Deceptive strata.

3. Method

3.1. Source Datasets and Sampling

We draw samples from four publicly available VQA benchmarks chosen to span a wide range of visual domains and question styles: **VQAv2** [5] (general everyday scenes, 500 samples from the validation split); **TextVQA** [11] (text recognition in natural images, 300 from the validation split); **ChartQA** [9] (chart and figure understanding, 500 from the validation split); and **RealWorldQA** [16] (real-world photographs, 200 from the test set). The total corpus is **1,500** image-question pairs.

3.2. VLM Ensemble

Meaningful disagreement requires architectural diversity; a set of nearly identical models will converge on similar errors. We select four open-source VLMs spanning a wide range of visual encoders and language backbones (Table 1): Qwen2-VL-7B-Instruct [15] (dynamic-resolution ViT + Qwen2); LLaVA-1.6-Mistral-7B [7] (CLIP ViT-L + Mistral-7B); InternVL2-8B [2] (InternViT-300M + InternLM2-7B); and MiniCPM-V 2.6 [17] (SigLIP + MiniCPM-3B). All models run at bfloat16 precision on A100-40GB GPUs with no quantisation. Each model answers each question independently with greedy decoding (`max_new_tokens=64`, `do_sample=False`). No brevity instruction is prepended; each model receives the raw question string paired with the image, using its default chat template.

Table 1. VLM ensemble used for disagreement scoring.

Tag	Model	Visual enc.	LM backbone
`qwen`	Qwen2-VL-7B	Dynamic ViT	Qwen2-7B
`llava`	LLaVA-1.6-M-7B	CLIP ViT-L	Mistral-7B
`internvl`	InternVL2-8B	InternViT-300M	InternLM2-7B
`minicpm`	MiniCPM-V 2.6	SigLIP	MiniCPM-3B

3.3. Disagreement Score

Given four predictions p_1, p_2, p_3, p_4 for a sample, we compute the disagreement score D as follows.

Step 1: normalisation. Each prediction is lowercased, punctuation-stripped, and article-removed following the standard VQA normalisation [1].

Step 2: semantic clustering. Normalised predictions are first grouped by exact string match. We then merge groups whose representative sentence-BERT embeddings [10] exceed cosine similarity ≥ 0.8, handling semantically equivalent paraphrases (*e.g.* “automobile” and “car”). A threshold sweep at $\{0.70, 0.85, 0.90\}$ preserves 84.8% of stratum labels at ± 0.05 from our 0.80 default, confirming label stability.

Step 3: disagreement score. Let $k^* = \max_j |C_j|$ be the size of the largest prediction cluster.

$$D = 1 - \frac{k^*}{4}. \tag{1}$$

$D = 0$ means all four models agree; $D = 0.75$ means all four give different answers.

Stratum assignment. Each sample is additionally scored for correctness against the ground truth using VQA soft accuracy [1] (for VQAv2 and TextVQA, both of which provide multiple human annotations) or normalised exact match (for ChartQA and RealWorldQA), yielding four strata:

- **Easy** ($D = 0$, all correct): saturated; minimal benchmark value.
- **Deceptive** ($D = 0$, all wrong): unanimous systematic error; reveals architectural blind spots.
- **Medium** ($D = 0.25$, 3-of-4 agree): contested; one model diverges.
- **Hard** ($D \geq 0.5$, no 3+ consensus): high information; models genuinely disagree.

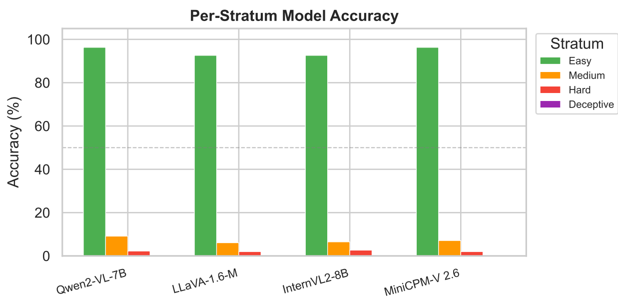

Figure 1. **Per-stratum model accuracy.** Easy samples are answered correctly by nearly all models (>92%); Hard samples see accuracy collapse to 2–3%, confirming disagreement as a difficulty signal.

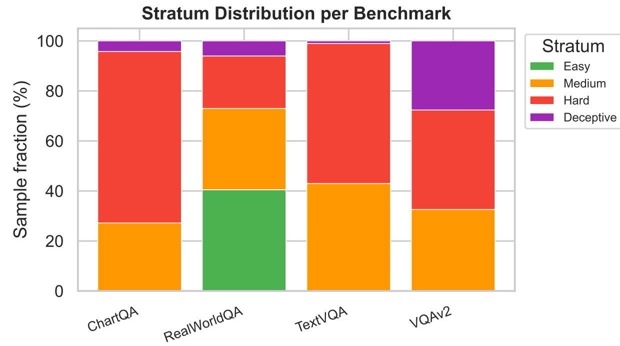

Figure 2. **Stratum distribution per benchmark.** RealWorldQA has the highest Easy fraction (40.5%); ChartQA and TextVQA are dominated by Hard samples. VQAv2's large Deceptive fraction (27.6%) reflects systematic answer-format mismatch between model outputs and the short-form annotation convention.

Table 2. Per-source-dataset stratum statistics.

Dataset	N	Easy%	Medium%	Hard%	Deceptive%
ChartQA	500	0.0	27.2	68.6	4.2
RealWorldQA	200	40.5	32.5	21.0	6.0
TextVQA	300	0.0	43.0	56.0	1.0
VQAv2	500	0.0	32.6	39.8	27.6

Table 3. Accuracy (%) per model on each stratum.

Model	Easy	Medium	Hard	Deceptive
Qwen2-VL-7B	96.3	9.1	2.3	0.0
LLaVA-1.6-M	92.6	6.1	2.0	0.0
InternVL2-8B	92.6	6.5	2.7	0.0
MiniCPM-V 2.6	96.3	7.1	2.0	0.0

3.4. Scale-Based Oracle Validation

To confirm that high-disagreement and unanimous-but-wrong strata reflect genuine task difficulty—rather than idiosyncratic failure modes of the four 7B-scale ensemble members—we run Qwen2-VL-72B-Instruct [15] ($\approx 10\times$ larger) on 455 items: 81 Easy, 100 Medium, 100 Hard, and all 174 Deceptive samples. Using a same-family model at much larger scale is a *stronger* argument than cross-family agreement: if the Hard stratum remains harder for a 72B model sharing pre-training lineage with one ensemble member, the difficulty is structural rather than a product of architectural disagreement. The 72B model is loaded with 4-bit NF4 quantisation [3] on an A100-80GB GPU.

4. Results

4.1. Stratum Distribution

Table 2 reports the fraction of each stratum per source dataset. Across the full corpus, **5.4%** of samples fall in the Easy stratum (*i.e.* they have been effectively "solved" by all four VLMs). The saturated fraction varies substantially across datasets: RealWorldQA has the highest Easy fraction (**40.5%**), suggesting that presence/absence and coarse attribute questions in real-world photographs are largely mastered at 7B scale. In contrast, VQAv2, TextVQA, and ChartQA have **0%** Easy samples, indicating that none of the 1,300 probed questions on these three benchmarks were answered correctly by all four models unanimously—even for VQAv2, where models systematically generate verbose answers that fail normalized exact match against the expected short-form human annotations.

The Deceptive stratum accounts for **11.6%** of all samples—of which **27.6%** of VQAv2 samples fall in this stratum, by far the highest rate. Manual inspection of 25 random Deceptive samples reveals two causes: **(a) answer-format mismatch**, where models produce verbose, semantically correct responses (*e.g.* "Yes, there is a red sandal in the image.") that fail the short-form VQAv2 convention ("yes"); and **(b) genuine shared visual errors** such as systematic miscounts or fine-grained colour confusions across all four architectures. Case (a) dominates in VQAv2; the oracle analysis in Section 4.3 shows case (b) dominates elsewhere.

Figure 2 shows the stacked stratum distribution per dataset. ChartQA has the highest Hard fraction (**68.6%**), confirming that chart-reading requires specialised visual reasoning not yet solved at 7B scale.

4.2. Per-Model Accuracy per Stratum

Table 3 reports individual model accuracy on each stratum. Easy accuracy ranges from 92.6% to 96.3% across models—near-perfect but below 100%, as sentence-BERT-based clustering can group semantically similar predictions that still differ from the normalized ground-truth string. On Hard samples, per-model accuracy collapses to **2.0%–2.7%**, with InternVL2-8B performing marginally better than the other ensemble members. The Deceptive stratum yields 0% ac-

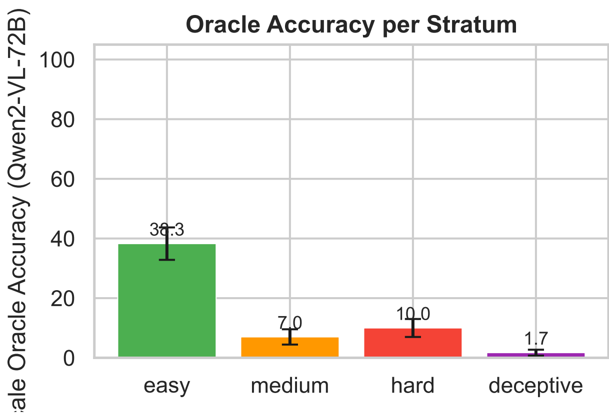

Figure 3. **Qwen2-VL-72B accuracy per stratum (oracle validation, $n = 455$).** Easy: 38.3%; Medium: 7.0%; Hard: 10.0%; Deceptive: 1.7%. The Easy–Deceptive gap of **36.6** pp is the strongest evidence that multi-model disagreement and unanimous-but-wrong consensus both reflect genuine difficulty rather than ensemble-specific idiosyncrasies.

curacy for every model—by construction, since Deceptive requires all four predictions to cluster together while failing the ground truth.

Figure 1 visualises these results across models and strata.

4.3. Qwen2-VL-72B Oracle Validation

Figure 3 shows Qwen2-VL-72B accuracy per stratum. Easy: **38.3%**; Medium: **7.0%**; Hard: **10.0%**; Deceptive: **1.7%** (3 of 174 items). The Easy–Hard gap is **28.3** pp and the Easy–Deceptive gap is **36.6** pp, both well above our 15 pp validity threshold. This confirms that multi-model disagreement is not a spurious signal but correlates with genuine task difficulty as measured by an independent, stronger oracle. (The 3 pp Hard–Medium difference is not statistically significant, $\chi^2 = 0.26$, $p = 0.61$.)

Deceptive is the hardest stratum for the oracle. The 72B oracle solves only 1.7% of Deceptive items (95% CI [0.0%, 3.7%]), substantially below even the Hard stratum. Per source: 0/138 VQAv2, 0/21 ChartQA, 0/3 TextVQA, 3/12 RealWorldQA. The VQAv2 zero is partly a verbosity artefact; the ChartQA zero is genuine—a same-family 72B model also fails the chart questions that confuse all four 7B members, supporting the Deceptive stratum as a window onto pretraining-level blind spots. Because all 81 Easy items come from RealWorldQA, the Easy oracle accuracy (38.3%) and the Easy–Hard gap also pick up a source-dataset shift (see Limitations).

4.4. Disagreement by Question Type

Figure 4 shows the stratum distribution across seven question types (counting, spatial, attribute, relational, existence, chart-reading, and other).

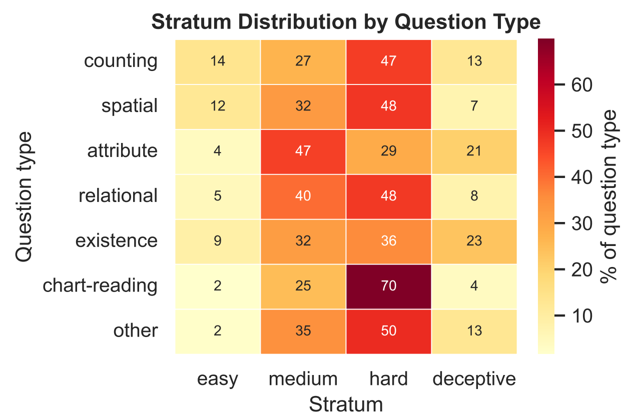

Figure 4. **Stratum × question type.** Chart-reading and counting questions are disproportionately Hard; attribute questions are the least hard.

Chart-reading and counting questions have the highest Hard fractions (**69.7%** and **47.0%**), consistent with prior work showing that enumerative and quantitative visual reasoning challenges current VLMs disproportionately [9, 14]. Attribute questions are the least hard (Hard fraction 28.8%), suggesting that coarse visual attribute detection (*e.g.* colour, shape) is more reliably solved at 7B scale. The chi-square test for association between stratum and question type is highly significant ($\chi^2 = 123.2$, $p < 0.001$, $df = 18$).

5. Discussion, Release, and Conclusion

Multi-model disagreement is a cheap, robust signal of VQA difficulty: four open-source 7B VLMs stratify 1,500 samples into informative buckets validated by a 72B same-family oracle, at zero annotation cost. Re-evaluating models on the Hard stratum alone would roughly halve evaluation compute while providing *more* discriminative signal; we advocate routine stratum-stratified reporting over aggregate accuracy. The Deceptive stratum—which the 72B oracle solves at only 1.7%—is the most diagnostic of shared pretraining-level failures and warrants mining for label auditing. We release VQA-DISAGREE, a 456-sample stratified subset (up to 125 per stratum, Easy capped at 81) including the source image, question, ground truth, four ensemble predictions, disagreement score, stratum label, and 72B oracle result for the validated 455-item subset. Dataset: `huggingface.co/datasets/AshwinP10/VQA-Disagree`; code: `github.com/AshwinP10/vqa-disagree`.

Limitations. All 81 Easy items come from RealWorldQA, so the 28.3 pp Easy–Hard oracle gap partly reflects a source-dataset shift. Our 7B ensemble may share failure modes invisible at frontier scale, and the rule-based classifier assigns 50.9% of samples to “other,” limiting question-type resolution.

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